Conductivity at conical crossings

Abstract

We consider independent electrons in a periodic crystal in their ground state, and turn on a uniform electric field at some prescribed time. We rigorously define the current per unit area and study the static conductivity of a two-dimensional semimetal. The conductivity tensor is isotropic and each conical intersection contributes as $\frac{1}{16}$ to the total conductivity.

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1 Introduction

We consider a two-dimensional perfect crystal with periodic lattice $\mathcal{R} \subset \mathbb{R}^2$. We use atomic units and a spinless model, with Hamiltonian

$$H = -\frac{1}{2}\Delta + V \quad \text{on } L^2(\mathbb{R}^2;\mathbb{C}), \tag{1}$$

where $V \in H^1_{\mathrm{per}}(\Omega;\mathbb{R})$ and Ω is a unit cell of $\mathcal{R}$. At zero temperature, the ground-state density matrix is given by

$$\gamma_0 = \mathbb{1}(H \leq \mu_{\mathrm{F}}), \tag{2}$$

where $\mu_{\mathrm{F}} \in \mathbb{R}$ is the Fermi level.

The purpose of this article is to analyze mathematically the behavior of the current appearing in the crystal when a uniform external field is turned on instantaneously at the initial time $t = 0$. Fix $e_\beta, e_\alpha \in \mathbb{R}^2$ once and for all, for the field and observation directions respectively. They need not be normalized, linearly independent, or orthogonal. Put $x_\beta = x \cdot e_\beta$ and $p = -i\nabla$. For $\varepsilon \geq 0$,

$$H_\varepsilon = H - \varepsilon x_\beta, \tag{3}$$

where H_ε denotes the self-adjoint closure of its restriction to $C_c^\infty(\mathbb{R}^2; \mathbb{C})$ (see Proposition 2.2). The state at $t \geq 0$ is

$$\gamma_\varepsilon(t) = e^{-itH_\varepsilon} \gamma_0 e^{itH_\varepsilon}. \tag{4}$$

The current is defined as

$$j_\varepsilon(t) = \underline{\mathrm{Tr}}(J\gamma_\varepsilon(t)), \tag{5}$$

where $\underline{\mathrm{Tr}}$ is the trace per unit area, defined in Section 2.1, and

$$J = p \cdot e_\alpha. \tag{6}$$

For time-reversible 2D semimetals such as graphene, the time-averaged conductivity

$$\sigma = \lim_{t\to\infty} \frac{1}{t} \int_0^t \lim_{\varepsilon\to 0^+} \frac{j_\varepsilon(t')}{\varepsilon}\, dt' \tag{7}$$

has a finite value equal to $\frac{1}{16} e_\alpha \cdot e_\beta$ times the number of Dirac points in the Brillouin zone.

Note that our results use an averaging in time, and we are unable to conclude anything about what would be the naive definition of the conductivity

$$\lim_{t\to\infty} \lim_{\varepsilon\to 0^+} \frac{j_\varepsilon(t)}{\varepsilon}. \tag{8}$$

A form of averaging of time fluctuations is always necessary to infer zero-frequency behavior from step responses in non-dissipative systems, even in the linear case. The easiest way to see this is by the very simple model for the relationship between an input $I(t)$ and an output $O(t)$:

$$i\dot{O}(t) = \omega O(t) + I(t). \tag{9}$$

This simplified model describes a forced oscillator with eigenfrequency ω, and arises from the linear response of the time-dependent Schrödinger equation of a two-level system. For a constant input I_0, there is a steady state solution $O_0 = \widehat{R}_0 I_0$, where $\widehat{R}_0 = -\frac{1}{\omega}$ is the zero-frequency transfer function of the system. However, since this system is oscillatory, this steady state may never be reached: if I is brutally switched on at time 0 with $I(t) = O(t) = 0$, if $t \leq 0$, $I(t) = I_0$ if $t > 0$, then $O(t) = O_0(1 - \mathrm{e}^{-i\omega t}) = \widehat{R}_0 I_0 (1 - \mathrm{e}^{-i\omega t})$ and we cannot define $\widehat{R}_0$ as the limit of $O(t)/I_0$ when t goes to infinity. However, by averaging we obtain

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t \frac{O(t')}{I_0}\, dt' = \widehat{R}_0.$$

In our results in the linear response regime, we consider the quantity

$$j^{\mathrm{LR}}(t) = \lim_{\varepsilon\to 0^+} \frac{j_\varepsilon(t)}{\varepsilon}.$$

and then compute the conductivity in the limit $t \to \infty$. This order of limits is necessary to ensure that the electronic state never deviates significantly from its equilibrium. Considering the opposite limit, i.e. the infinite-time dynamics of $j_\varepsilon(t)$ at ε fixed, is an extremely hard problem, as it is for any dynamics of non-dissipative systems. Our method of proof is based on the standard gauge change $\widetilde{\psi}(x,t) = e^{-i\varepsilon t x_\beta} \psi(x,t)$ that transforms the constant in time but non-spatially-periodic Hamiltonian H_ε into the time-dependent Hamiltonian $\widetilde{H}_\varepsilon(t) = \frac{1}{2}(p + \varepsilon t e_\beta)^2 + V$. This Hamiltonian is spatially periodic, and the study of its dynamics can be reduced via Bloch–Floquet theory to that of its fibers

$H_{k+\varepsilon t e_\beta}$ acting on periodic functions (Section 3). Fiber by fiber, this time-dependent Hamiltonian can then be treated using the tools of time-dependent perturbation theory (Section 4). For each fixed t, we use linear response to expand $j_\varepsilon(t)$ to first order in ε (Proposition 4.1).

The structure of the paper is as follows. We describe our results in Section 2: we define the current in Proposition 2.2, and state Theorem 2.4. We devote Section 3 to preliminaries about the regularity and Bloch decomposition of the current. Section 4 states and proves the linear response result. Section 5 is devoted to the proof of the conductivity formula. Finally two short Appendices are devoted to technical issues.

2 Main results: electrical current in periodic materials

2.1 Notation

We denote by $\mathcal{R}^*$ the dual lattice

$$\mathcal{R}^* = \{K \in \mathbb{R}^2 : K \cdot R \in 2\pi\mathbb{Z} \text{ for all } R \in \mathcal{R}\},$$

by Ω a unit cell in physical space, and by $\mathcal{B}$ a unit cell in reciprocal space, called the Brillouin zone. We write $\partial_\alpha = e_\alpha \cdot \nabla_k$ and $\partial_\beta = e_\beta \cdot \nabla_k$ for the two fixed directional derivatives.

The *$\mathcal{R}$-periodic Lebesgue and Sobolev spaces* are denoted by

$$L^p_{\mathrm{per}} := \left\{u \in L^p_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{C}) \mid u \ \mathcal{R}\text{-periodic}\right\},$$
$$H^s_{\mathrm{per}} := \left\{u \in H^s_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{C}) \mid u \ \mathcal{R}\text{-periodic}\right\}.$$

The space of bounded linear operators on a Hilbert space $\mathcal{H}$ is denoted by $\mathcal{L}(\mathcal{H})$, and the Schatten class of bounded operators $A \in \mathcal{L}(\mathcal{H})$ such that $\mathrm{Tr}(|A|^p) < \infty$ by $\mathfrak{S}_p(\mathcal{H})$. For $R \in \mathcal{R}$, we denote by τ_R the translation operator formally defined by $\tau_R\phi = \phi(\cdot - R)$. Depending on the context, τ_R will be seen as a unitary operator on $L^2(\mathbb{R}^2;\mathbb{C})$, or as a linear operator on some $\mathcal{R}$-translation invariant subspace of $\mathcal{D}'(\mathbb{R}^2;\mathbb{C})$. A bounded operator on $L^2(\mathbb{R}^2;\mathbb{C})$ is called $\mathcal{R}$-periodic if it commutes with τ_R for all $R \in \mathcal{R}$. An unbounded self-adjoint operator on $L^2(\mathbb{R}^2;\mathbb{C})$ is called $\mathcal{R}$-periodic if its resolvent is $\mathcal{R}$-periodic. A bounded $\mathcal{R}$-periodic operator $A \in \mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))$ is called locally trace-class if $\chi A \chi \in \mathfrak{S}_1(L^2(\mathbb{R}^2;\mathbb{C}))$ for any compactly supported function $\chi \in L^\infty(\mathbb{R}^2;\mathbb{C})$. For $p \geq 1$, we denote by $\mathfrak{S}_{p,\mathrm{per}}$ the space of $\mathcal{R}$-periodic operators $A \in \mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))$ such that $|A|^p$ is locally trace class. Any operator $A \in \mathfrak{S}_{1,\mathrm{per}}$ has a density $\rho_A \in L^1_{\mathrm{per}}$ characterized by

$$\forall \chi \in C^\infty_{\mathrm{c}}(\mathbb{R}^2;\mathbb{C}), \quad \mathrm{Tr}(A\chi) = \int_{\mathbb{R}^2} \rho_A \chi.$$

The trace per unit area of an operator $A \in \mathfrak{S}_{1,\mathrm{per}}$ is defined as

$$\underline{\mathrm{Tr}}(A) = \frac{1}{|\Omega|}\,\mathrm{Tr}_{L^2(\mathbb{R}^2;\mathbb{C})}(\mathbb{1}_\Omega A \mathbb{1}_\Omega) = \fint_\Omega \rho_A,$$

where $\mathbb{1}_\Omega$ is the characteristic function of the unit cell Ω, and $\fint_\Omega$ is a shorthand notation for $\frac{1}{|\Omega|}\int_\Omega$. This formula is independent of the choice of the unit cell Ω.

Since we are dealing here with periodic materials, we will use the Bloch transform (also called Bloch-Floquet transform) [2]. For $K \in \mathcal{R}^*$, let T_K be the unitary multiplication operator on L^2_{per} defined by

$$\forall v \in L^2_{\mathrm{per}}, \quad (T_K v)(x) = \mathrm{e}^{-iK\cdot x} v(x) \quad \text{for a.a. } x \in \mathbb{R}^2,$$

and

$$L^2_{\mathrm{qp}}(L^2_{\mathrm{per}}) := \left\{\mathbb{R}^2 \ni k \mapsto u_k \in L^2_{\mathrm{per}} \mid \int_{\mathcal{B}} \|u_k\|^2_{L^2_{\mathrm{per}}}\, dk < \infty,\ u_{k+K} = T_K u_k \text{ for all } K \in \mathcal{R}^* \text{ and a.a. } k \in \mathbb{R}^2\right\},$$

the Hilbert space of $\mathcal{R}^*$-quasi-periodic $L^2_{\rm per}$-valued functions on $\mathbb{R}^2$ endowed with the inner product

$$\langle u, v\rangle_{L^2_{\rm qp}(L^2_{\rm per})} = \fint_{\mathcal{B}} \langle u_k, v_k\rangle_{L^2_{\rm per}}\, dk.$$

Here and below, the subscript qp refers to the quasi-periodicity property. The Bloch transform then is the unitary map from $L^2(\mathbb{R}^2;\mathbb{C})$ to $L^2_{\rm qp}(L^2_{\rm per})$ defined for $u \in C^\infty_{\rm c}(\mathbb{R}^2;\mathbb{C})$ by

$$\forall k \in \mathbb{R}^2, \quad \forall x \in \mathbb{R}^2, \quad u_k(x) = \sum_{R\in\mathcal{R}} u(x+R)\, {\rm e}^{-ik\cdot(x+R)}. \tag{10}$$

Its inverse is given by

$$u(x) = \fint_{\mathcal{B}} {\rm e}^{ik\cdot x} u_k(x)\, dk, \quad \text{for a.a. } x \in \mathbb{R}^2. \tag{11}$$

Any $\mathcal{R}$-periodic operator $A \in \mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))$ is decomposed by the Bloch transform in the sense that there exists a function $k \mapsto A_k$ in $L^\infty_{\rm qp}(\mathcal{L}(L^2_{\rm per}))$ such that for any $u \in L^2(\mathbb{R}^2;\mathbb{C})$ and almost all $k \in \mathbb{R}^2$, $(Au)_k = A_k u_k$, and

$$A_{k+K} = T_K A_k T_K^*, \quad \text{for all } K \in \mathcal{R}^* \text{ and a.a. } k \in \mathbb{R}^2. \tag{12}$$

The A_k's are called the fibers of the operator A. If $A \in \mathfrak{S}_{1,\rm per}$, then the function $k \mapsto A_k$ is in $L^1_{\rm qp}(\mathfrak{S}_1(L^2_{\rm per}))$, the function $k \mapsto {\rm Tr}(A_k)$ is in $L^1_{\rm loc}(\mathbb{R}^2)$, $\mathcal{R}^*$-periodic, and we have

$$\underline{\rm Tr}(A) = (2\pi)^{-2} \int_{\mathcal{B}} {\rm Tr}(A_k)\, dk.$$

The Bloch decomposition theorem can be extended to unbounded $\mathcal{R}$-periodic self-adjoint operators using the resolvent [2].

In the case of the periodic Hamiltonian operator H given by (1), we have

$$H_k = \frac{1}{2}(-i\nabla + k)^2 + V. \tag{13}$$

For each $k \in \mathbb{R}^2$, H_k is a bounded below self-adjoint operator on $L^2_{\rm per}$ with domain $H^2_{\rm per}$ and compact resolvent. Let $(\lambda_{n,k})_{n\in\mathbb{N}^*}$ be the non-decreasing sequence of eigenvalues of H_k counting multiplicities

$$\lambda_{1,k} \le \lambda_{2,k} \le \lambda_{3,k} \le \cdots, \quad \lim_{n\to\infty} \lambda_{n,k} = +\infty,$$

and we use the convention $\lambda_{0,k} = -\infty$. We denote by $(u_{n,k})_{n\in\mathbb{N}^*} \in (H^2_{\rm per})^{\mathbb{N}^*}$ an $L^2_{\rm per}$-orthonormal basis of associated eigenfunctions:

$$H_k u_{n,k} = \lambda_{n,k} u_{n,k}, \quad \langle u_{m,k}, u_{n,k}\rangle_{L^2_{\rm per}} = \delta_{m,n}.$$

For $n \ge 1$ and $k \in \mathbb{R}^2$, set

$$P_{n,k} = \mathbb{1}(H_k \le \lambda_{n,k}), \qquad P_{0,k} = 0.$$

Whenever $\lambda_{n,k} < \lambda_{n+1,k}$, $P_{n,k}$ is the spectral projector associated with the lowest n eigenvalues, counting multiplicities:

$$P_{n,k} = \sum_{m=1}^{n} |u_{m,k}\rangle\langle u_{m,k}|. \tag{14}$$

Since H_k is quasi-periodic, so is $P_{n,k}$, and the eigenvalues $\lambda_{n,k}$ are $\mathcal{R}^*$-periodic functions of k. By a min-max argument (see e.g. [2]), there exists $\underline{C_1}, \overline{C_1} \in \mathbb{R}$, and $\underline{C_2}, \overline{C_2} > 0$ such that

$$\underline{C_1} + \underline{C_2} n \le \lambda_{n,k} \le \overline{C_1} + \overline{C_2} n. \tag{15}$$

The ground-state density matrix γ_0 has Bloch fibers

$$P_k = \mathbb{1}(H_k \le \mu_{\rm F}). \tag{16}$$

Their ranks are uniformly bounded by the eigenvalue estimate above. The current operator J is also $\mathcal{R}$-periodic, with fibers

$$J_k = (p+k)\cdot e_\alpha = \partial_\alpha H_k.$$

Lastly, for each $q \in \mathbb{R}^2$, we denote by G_q the unitary multiplication operator on $L^2(\mathbb{R}^2;\mathbb{C})$ defined by

$$\forall u \in L^2(\mathbb{R}^2;\mathbb{C}), \quad (G_q u)(x) = \mathrm{e}^{iq\cdot x}u(x) \quad \text{for a.a. } x \in \mathbb{R}^2. \tag{17}$$

The operator G_q is not $\mathcal{R}$-periodic, except when $q \in \mathcal{R}^*$ (in which case G_q is fibered, with $G_{q,k} = T_{-q}$ for all k). However, for any $\mathcal{R}$-periodic operator $A \in \mathcal{L}(L^2(\mathbb{R}^2;\mathbb{C}))$ and any $q \in \mathbb{R}^2$, the operator $G_q A G_q^*$ is $\mathcal{R}$-periodic and its Bloch decomposition is given by

$$(G_q A G_q^*)_k = A_{k-q}, \quad \text{for a.a. } k \in \mathbb{R}^2. \tag{18}$$

2.2 The Bloch theorem

Before attacking the well-posedness of the current $j_\varepsilon(t) = \underline{\mathrm{Tr}}(J\gamma_\varepsilon(t))$ for $\varepsilon, t \neq 0$, we first study an easier special case.

Proposition 2.1 (Bloch theorem)**.** *The current satisfies*

$$\begin{aligned} &j_0(t) = 0, \quad \forall t \geq 0 \quad \textit{(no current in the absence of external field)},\\ &j_\varepsilon(0) = 0, \quad \forall \varepsilon \geq 0 \quad \textit{(continuity of the current at } t = 0\textit{)}. \end{aligned}$$

Proof. We have

$$j_0(t) = j_\varepsilon(0) = \frac{1}{4\pi^2}\int_{\mathcal{B}} \mathrm{Tr}(\partial_\alpha H_k P_k)\, dk.$$

Since V is real, $H_{-k} = \overline{H_k}$ and $P_{-k} = \overline{P_k}$, where the bar denotes complex conjugation. Moreover, $\partial_\alpha H_{-k} = -\overline{\partial_\alpha H_k}$. Thus the real-valued, $\mathcal{R}^*$-periodic function $k \mapsto \mathrm{Tr}(\partial_\alpha H_k P_k)$ is odd, and its integral vanishes. □

2.3 Definition of the current

For $\varepsilon > 0$, the operator $H_\varepsilon = H - \varepsilon x_\beta$ is not $\mathcal{R}$-periodic, and we would naively expect that the density matrix $\gamma_\varepsilon(t)$ is not either. Yet, this operator is in fact $\mathcal{R}$-periodic. Physically, this is due to the fact that although $-\varepsilon x \cdot e_\beta$ is not periodic, its negative gradient is constant, hence periodic. The proof relies on the standard gauge transform

$$\widetilde{\psi}(x,t) = G_{-\varepsilon t e_\beta}\psi(x,t) = e^{-i\varepsilon t x_\beta}\psi(x,t), \tag{19}$$

where G_q was defined in (17), and the gauge-transformed operators

$$U_\varepsilon(t,t') = G_{-\varepsilon t e_\beta} e^{-i(t-t')H_\varepsilon} G^*_{-\varepsilon t' e_\beta}, \tag{20}$$

and

$$\widetilde{\gamma}_\varepsilon(t) = G_{-\varepsilon t e_\beta}\gamma_\varepsilon(t) G^*_{-\varepsilon t e_\beta} = U_\varepsilon(t)\gamma_0 U_\varepsilon(t)^*, \tag{21}$$

where $U_\varepsilon(t) = U_\varepsilon(t,0)$. Through this change of gauge, the dynamics induced by H_ε is equivalent to that induced by the time-dependent periodic Hamiltonian

$$\widetilde{H}_\varepsilon(t) = G_{-\varepsilon t e_\beta} H_\varepsilon G^*_{-\varepsilon t e_\beta} + \varepsilon x_\beta = \frac{1}{2}(p + \varepsilon t e_\beta)^2 + V. \tag{22}$$

Its Bloch fibers are

$$\widetilde{H}_{\varepsilon,k}(t) = \frac{1}{2}(p + k + \varepsilon t e_\beta)^2 + V = H_{k+\varepsilon t e_\beta}. \tag{23}$$

We sum up these arguments in the proposition below, together with elements that we shall use for defining the current.

Proposition 2.2. *Let $V \in H^1_{\mathrm{per}}(\Omega;\mathbb{R})$.*

1. *For every $\varepsilon \in \mathbb{R}$, H_ε is essentially self-adjoint on $C_c^\infty(\mathbb{R}^2;\mathbb{C})$ and therefore generates a unitary group on $L^2(\mathbb{R}^2;\mathbb{C})$.*

2. *The operator $\widetilde{H}_\varepsilon(t)$ is self-adjoint with domain $H^2(\mathbb{R}^2;\mathbb{C})$ and $\mathcal{R}$-periodic. The propagator $U_\varepsilon(t,t')$ is $\mathcal{R}$-periodic, with fibers $U_{\varepsilon,k}(t,t')$ solving*

$$i\partial_t U_{\varepsilon,k}(t,t') = H_{k+\varepsilon t e_\beta} U_{\varepsilon,k}(t,t'), \qquad U_{\varepsilon,k}(t',t') = \mathrm{Id}. \tag{24}$$

3. *For $t \geq 0$, $J\gamma_\varepsilon(t) \in \mathfrak{S}_{1,\mathrm{per}}$, and*

$$\begin{aligned} j_\varepsilon(t) &= \frac{1}{4\pi^2}\int_{\mathcal{B}} \mathrm{Tr}\left(\partial_\alpha H_{k+\varepsilon t e_\beta}\widetilde{\gamma}_{\varepsilon,k}(t)\right)\, dk \\ &= \frac{1}{4\pi^2}\int_{\mathcal{B}} \mathrm{Tr}\left(\partial_\alpha H_{k+\varepsilon t e_\beta} U_{\varepsilon,k}(t) P_k U_{\varepsilon,k}(t)^*\right)\, dk. \end{aligned} \tag{25}$$

The results of Proposition 2.2 are proved in Section 3 for completeness. The situation can be summed up in the commutative diagrams of Figure 1.

$$\begin{array}{ccc} \gamma_0 & \xrightarrow{e^{-itH_\varepsilon}} & \gamma_\varepsilon(t) \\ \downarrow{\scriptstyle \mathrm{Id}} & & \downarrow{\scriptstyle G_{-\varepsilon t e_\beta}} \\ \gamma_0 & \xrightarrow{U_\varepsilon(t)} & \widetilde{\gamma}_\varepsilon(t) \end{array} \qquad \begin{array}{ccc} \{P_k\}_k & \longrightarrow & \{\gamma_{\varepsilon,k}(t)\}_k \\ \downarrow{\scriptstyle \mathrm{Id}} & & \downarrow{\scriptstyle k\mapsto k+\varepsilon t e_\beta} \\ \{P_k\}_k & \xrightarrow{\{U_{\varepsilon,k}(t)\}_k} & \{\widetilde{\gamma}_{\varepsilon,k}(t)\}_k \end{array}$$

Figure 1: Relations between the density matrices and their Bloch fibers. An arrow labelled by a unitary U denotes conjugation by U. The right lower arrow means $\widetilde{\gamma}_{\varepsilon,k}(t) = \gamma_{\varepsilon,k+\varepsilon t e_\beta}(t)$.

2.4 Semimetals

The position of the Fermi level in the band diagram is key to determining the electronic properties of the medium. We assume the following.

Assumption 2.3 (semimetal)**.** *There are $N, M \geq 1$ and distinct points $k_1, \ldots, k_M$ modulo $\mathcal{R}^*$ such that*

$$\lambda_{N,k} < \mu_{\mathrm{F}} < \lambda_{N+1,k} \qquad (k \notin \{k_1,\ldots,k_M\} \bmod \mathcal{R}^*).$$

For each $i = 1, \ldots, M$, the two bands meet at an isolated conical crossing:

$$\lambda_{N-1,k_i} < \lambda_{N,k_i} = \mu_{\mathrm{F}} = \lambda_{N+1,k_i} < \lambda_{N+2,k_i}, \tag{26}$$

$$\lambda_{N,k} = \mu_{\mathrm{F}} - v_i|k-k_i| + O(|k-k_i|^2), \tag{27}$$

$$\lambda_{N+1,k} = \mu_{\mathrm{F}} + v_i|k-k_i| + O(|k-k_i|^2), \tag{28}$$

where $v_i > 0$.

The Fermi surface is the finite set

$$\mathcal{S} = \{k_1,\ldots,k_M\} \bmod \mathcal{R}^*. \tag{29}$$

For $k \notin \mathcal{S}$, $P_k = P_{N,k}$ has rank N. Its values at the crossings do not affect any Brillouin-zone integral. Since V is real, $H_{-k} = \overline{H_k}$: the system has time-reversal symmetry. The passage to linear response is justified in Proposition 4.1.

2.5 Conductivity

In the following, $O(f(\varepsilon,t))$ denotes a quantity bounded by $Cf(\varepsilon,t)$, where C may depend on the potential, the Fermi level and the fixed directions, but not on t or ε. A subscript T allows the constant to depend on a fixed time horizon T. We write $o_T(\varepsilon)$ for a remainder whose quotient by ε tends uniformly to zero on that interval.

Theorem 2.4. *Under Assumption 2.3, the iterated limit exists and*

$$\sigma = \lim_{T\to\infty} \frac{1}{T} \int_0^T \left(\lim_{\varepsilon\to 0^+} \frac{j_\varepsilon(t)}{\varepsilon} \right) dt = \frac{M}{16}\, e_\alpha \cdot e_\beta.$$

The value of the conductivity is universal: it does not depend on the characteristics of the Hamiltonian, but only on the number of conical crossings. The conductivity tensor is isotropic and each conical intersection contributes 1/16.

3 Bloch decomposition and regularity of the current

In this section, we prove Proposition 2.2. We first point out an alternative interpretation of the gauge change. Formally, $\gamma_\varepsilon(t)$ satisfies

$$i\partial_t \gamma_\varepsilon = [H, \gamma_\varepsilon] - \varepsilon[x_\beta, \gamma_\varepsilon].$$

The commutator $[x_\beta, \gamma_\varepsilon]$ is $\mathcal{R}$-periodic, with fibers $i\partial_\beta \gamma_{\varepsilon,k}$. Therefore the fibers satisfy

$$i\partial_t \gamma_{\varepsilon,k} + i\varepsilon\partial_\beta \gamma_{\varepsilon,k} = [H_k, \gamma_{\varepsilon,k}] = L_{H_k}\gamma_{\varepsilon,k},$$

where $L_{H_k} = [H_k, \cdot]$ is the Liouvillian associated with H_k (see Section 4.1). The left-hand side is a linear advection equation, which suggests the method of characteristics: setting

$$\widetilde{\gamma}_{\varepsilon,k}(t) = \gamma_{\varepsilon,k+\varepsilon t e_\beta}(t), \tag{30}$$

we obtain

$$i\partial_t \widetilde{\gamma}_{\varepsilon,k}(t) = [H_{k+\varepsilon t e_\beta}, \widetilde{\gamma}_{\varepsilon,k}(t)].$$

The gauge transform $G_{-\varepsilon t e_\beta}$, equivalent to the change of variable (30), makes these remarks rigorous. We now prove Proposition 2.2.

Proof of the first assertion. The essential self-adjointness of H_ε follows from an extension of the Faris-Lavine theorem [1, Theorem X.38]. Let $\mathcal{C} = C_c^\infty(\mathbb{R}^2;\mathbb{C})$ be the set of infinitely differentiable, compactly supported functions.

Lemma 3.1 (Faris-Lavine criterion)**.** *Let V and W be real-valued measurable functions on $\mathbb{R}^2$, with $W \in L^2_{\mathrm{loc}}(\mathbb{R}^2;\mathbb{R})$. Suppose that*

1. *there exist $c > 0$ and $f \geq 0$ such that $W(x) \geq -c|x|^2 - f$, for a.a. $x \in \mathbb{R}^2$;*
2. *$-\frac{1}{2}\Delta + V + W + 2c|x|^2$ is essentially self-adjoint on $\mathcal{C}$;*
3. *for some $a < 1$, $-\frac{a}{2}\Delta + V$ is bounded below on $\mathcal{C}$.*

Then $-\frac{1}{2}\Delta + V + W$ is essentially self-adjoint on $\mathcal{C}$.

The proof of the above lemma is postponed until Appendix A.1.

We apply Lemma 3.1 with $W = -\varepsilon x_\beta$ and $c = 1$, noting that

$$-\varepsilon x \cdot e_\beta \geq -|x|^2 - \frac{\varepsilon^2 |e_\beta|^2}{4}.$$

In two dimensions, $V \in L^2_{\mathrm{per}}$ is infinitesimally bounded with respect to $-\Delta$, hence also with respect to the harmonic oscillator $-\frac{1}{2}\Delta + 2|x|^2$. The linear potential $-\varepsilon x_\beta$ is likewise infinitesimally bounded

with respect to this oscillator. The Kato–Rellich theorem therefore gives the essential self-adjointness of $-\frac{1}{2}\Delta + V - \varepsilon x_\beta + 2|x|^2$ on the core $\mathcal{C}$. Also, $-\frac{a}{2}\Delta + V$ is bounded below for every $a > 0$. Lemma 3.1 gives that H_ε is essentially self-adjoint on $\mathcal{C}$, and its propagator is the unitary group $(e^{-itH_\varepsilon})_{t\in\mathbb{R}}$.

Proof of the second assertion. The self-adjointness of

$$\widetilde{H}_\varepsilon(t) = \frac{1}{2}(p + \varepsilon t e_\beta)^2 + V$$

on $H^2(\mathbb{R}^2;\mathbb{C})$ follows from the Kato–Rellich theorem. To show the well-posedness of the dynamics, it suffices to study its fibers $H_{k+\varepsilon t e_\beta}$. We use the following lemma on time-dependent perturbations of the free-particle Hamiltonian on L^2_{per}.

Lemma 3.2. *Let $H_0 := -\frac{1}{2}\Delta$ be the free-particle Hamiltonian on L^2_{per}, and a C^1 map $H_1 : [0,T] \to \mathcal{L}(H^2_{\mathrm{per}}, L^2_{\mathrm{per}})$, taking its values in the set of H_0-bounded self-adjoint operators on L^2_{per} with relative bound lower than 1, that is: there exist $0 < a < 1$ and $b > 0$ such that*

$$\forall t \in [0,T], \quad \forall \phi \in H^2_{\mathrm{per}}, \quad \|H_1(t)\phi\|_{L^2_{\mathrm{per}}} \leq a\|H_0\phi\|_{L^2_{\mathrm{per}}} + b\|\phi\|_{L^2_{\mathrm{per}}}. \tag{31}$$

Then, for all $t \in [0,T]$, the operator defined by $H(t) = H_0 + H_1(t)$ is self-adjoint on L^2_{per} with domain H^2_{per}, and there exists a unique unitary propagator $(\mathcal{U}(t))_{t\in[0,T]}$ on L^2_{per} such that for $t \in [0,T]$, and $\phi_0 \in H^2_{\mathrm{per}}$, $\phi : t \mapsto \mathcal{U}(t)\phi_0$ is in $C([0,T];H^2_{\mathrm{per}}) \cap C^1([0,T];L^2_{\mathrm{per}})$, and solves the time-dependent Schrödinger equation

$$i\partial_t\phi(t) = H(t)\phi(t), \qquad \phi(0) = \phi_0.$$

The proof of the above lemma is postponed to Appendix A.2.

For $t \in [0,T]$, write $H_{k+\varepsilon t e_\beta} = H_0 + H_1(t)$, with

$$H_1(t) = (k + \varepsilon t e_\beta)\cdot p + \frac{1}{2}|k + \varepsilon t e_\beta|^2 + V.$$

The Sobolev embedding $H^2_{\mathrm{per}} \subset L^\infty_{\mathrm{per}}$ and $V \in L^2_{\mathrm{per}}$ imply the conditions of Lemma 3.2; the result follows.

Proof of the third assertion. We first compute the fibers of $\gamma_\varepsilon(t)$. Using (18), we have

$$\gamma_{\varepsilon,k}(t) = \left(G^*_{-\varepsilon t e_\beta} U_\varepsilon(t)\gamma_0 U_\varepsilon(t)^* G_{-\varepsilon t e_\beta}\right)_k \tag{32}$$

$$= U_{\varepsilon,k-\varepsilon t e_\beta}(t) P_{k-\varepsilon t e_\beta} U_{\varepsilon,k-\varepsilon t e_\beta}(t)^* \tag{33}$$

$$= \sum_{n:\, \lambda_{n,k-\varepsilon t e_\beta} \leq \mu_{\mathrm{F}}} |U_{\varepsilon,k-\varepsilon t e_\beta}(t)u_{n,k-\varepsilon t e_\beta}\rangle\langle U_{\varepsilon,k-\varepsilon t e_\beta}(t)u_{n,k-\varepsilon t e_\beta}|,$$

for almost every k. Since $u_{n,k} \in H^2_{\mathrm{per}}$, we deduce that

$$(J\gamma_\varepsilon(t))_k = \partial_\alpha H_k \gamma_{\varepsilon,k}(t)$$

is trace-class and finite-rank, with a uniform bound for $k \in \mathcal{B}$. The current is therefore well-defined. Since $k \mapsto \mathrm{Tr}(\partial_\alpha H_k \gamma_{\varepsilon,k}(t))$ is $\mathcal{R}^*$-periodic,

$$\begin{aligned} j_\varepsilon(t) &= \frac{1}{4\pi^2}\int_{\mathcal{B}} \mathrm{Tr}(\partial_\alpha H_k \gamma_{\varepsilon,k}(t))\, dk \\ &= \frac{1}{4\pi^2}\int_{\mathcal{B}} \mathrm{Tr}\big(\partial_\alpha H_{k+\varepsilon t e_\beta} \gamma_{\varepsilon,k+\varepsilon t e_\beta}(t)\big)\, dk, \end{aligned}$$

which gives (25).

4 Linear response

4.1 The Liouvillian and its partial inverse

In order to formulate and interpret our results, it is convenient to make use of the Liouvillian formalism. For each $k \notin \mathcal{S}$, the Liouvillian L_k associated with H_k is the commutator

$$L_k A = [H_k, A]. \tag{34}$$

The operator L_k is to density matrices what the Hamiltonian H_k is to pure states: its evolution is

$$e^{-itL_k} A = e^{-itH_k} A e^{itH_k}. \tag{35}$$

More precisely, L_k is the self-adjoint generator of this strongly-continuous unitary group on the Hilbert–Schmidt operators. Its action is easily understood in the eigenbasis of H_k:

$$L_k |u_{n,k}\rangle\langle u_{m,k}| = (\lambda_{n,k} - \lambda_{m,k}) |u_{n,k}\rangle\langle u_{m,k}|.$$

The operator L_k is not invertible. However, it is invertible on the block off-diagonal operators, i.e. operators A such that $P_k A P_k = (1 - P_k) A (1 - P_k) = 0$. Its partial inverse L_k^+ is given by

$$L_k^+ A = \sum_{n \le N < m} \frac{\langle u_{m,k}, A u_{n,k}\rangle |u_{m,k}\rangle\langle u_{n,k}| - \langle u_{n,k}, A u_{m,k}\rangle |u_{n,k}\rangle\langle u_{m,k}|}{\lambda_{m,k} - \lambda_{n,k}}. \tag{36}$$

The gap and the finite number of occupied states ensure convergence in Hilbert–Schmidt norm for bounded A. We extend L_k^+ to all bounded operators by setting it to zero on the diagonal blocks. In particular, $L_k L_k^+ A = A$ for off-diagonal A.

4.2 Spectral derivatives

For $k \notin \mathcal{S}$, the contour formula

$$P_k = \frac{1}{2\pi i} \oint_{\mathcal{C}} (z - H_k)^{-1} \, dz, \tag{37}$$

where $\mathcal{C}$ encloses the occupied eigenvalues, gives a locally analytic projector. Differentiating $[H_k, P_k] = 0$, we get

$$L_k \partial_\beta P_k = [P_k, \partial_\beta H_k].$$

Both sides are off-diagonal (for $\partial_\beta P_k$, differentiate $P_k^2 = P_k$). Hence

$$\partial_\beta P_k = L_k^+ [P_k, \partial_\beta H_k]. \tag{38}$$

4.3 Linear response

We now aim at obtaining an expansion of the current to first order in ε for a given t. This is a classical computation in response theory, sometimes known as the Kubo formula. We only need differentiability at $\varepsilon = 0$ on bounded time intervals; the estimates below do not require a lower bound on the gap.

Proposition 4.1. *For every $T > 0$, uniformly for $k \notin \mathcal{S}$ and $0 \le t \le T$,*

$$\begin{aligned} \operatorname{Tr}\left(\partial_\alpha H_{k+\varepsilon t e_\beta} U_{\varepsilon,k}(t) P_k U_{\varepsilon,k}(t)^*\right) &= \partial_\alpha \operatorname{Tr}(H_k P_k) + \varepsilon t \partial_\alpha \partial_\beta \operatorname{Tr}(H_k P_k) \\ &\quad - i\varepsilon \operatorname{Tr}\left(\partial_\alpha H_k (e^{-itL_k} - 1) L_k^+ \partial_\beta P_k\right) + o_T(\varepsilon). \end{aligned} \tag{39}$$

The current may be differentiated at $\varepsilon = 0$ under the Bloch integral. The limit defining $j^{\mathrm{LR}}(t)$ exists and may also be passed under any integral over a bounded time interval.

Proof. We first justify differentiation with the unbounded current observable. Fix $T > 0$ and $0 \le \varepsilon \le 1$. The common-domain propagator theorem used in Lemma 3.2 gives a uniform bound on $U_{\varepsilon,k}(t,s)$ as an operator on H^2_{per} for $0 \le s \le t \le T$ and $k \in \mathcal{B}$: on this compact parameter set, the graph norms and the C^1 bounds of the Hamiltonians are uniform. The occupied unit vectors also have uniformly bounded H^2_{per} norms, since their energies are bounded.

For such a vector u, set $\psi_\varepsilon(t) = U_{\varepsilon,k}(t)u$ and $\psi_0(t) = e^{-itH_k}u$. Since

$$H_{k+\varepsilon s e_\beta} - H_k = \varepsilon s \partial_\beta H_k + \frac{\varepsilon^2 s^2 |e_\beta|^2}{2},$$

Duhamel's formula gives

$$\psi_\varepsilon(t) - \psi_0(t) = -i \int_0^t e^{-i(t-s)H_k} \left(\varepsilon s \partial_\beta H_k + \frac{\varepsilon^2 s^2 |e_\beta|^2}{2} \right) \psi_\varepsilon(s)\, ds. \tag{40}$$

The difference is $O_T(\varepsilon)$ in L^2_{per} and bounded in H^2_{per}, hence $O_T(\sqrt{\varepsilon})$ in H^1_{per} by interpolation. Dividing (40) by ε therefore gives, in L^2_{per},

$$\frac{\psi_\varepsilon(t) - \psi_0(t)}{\varepsilon} \longrightarrow -i \int_0^t s\, e^{-i(t-s)H_k} \partial_\beta H_k e^{-isH_k} u\, ds.$$

All these bounds and the convergence are uniform in k, in occupied unit vectors u, and in $0 \le t \le T$.

Put $w = \psi_\varepsilon - \psi_0$. The identity

$$\langle \psi_\varepsilon, \partial_\alpha H_k \psi_\varepsilon \rangle - \langle \psi_0, \partial_\alpha H_k \psi_0 \rangle = 2\,\mathrm{Re}\langle w, \partial_\alpha H_k \psi_0 \rangle + \langle w, \partial_\alpha H_k w \rangle$$

has last term $O_T(\varepsilon^{3/2})$. Thus the current expectations are differentiable, with a uniform $o_T(\varepsilon)$ remainder. Summing over the N occupied vectors gives the density-matrix derivative

$$\frac{d}{d\varepsilon} \widetilde{\gamma}_{\varepsilon,k}(t)\Big|_{\varepsilon=0} = -i \int_0^t s\, e^{-i(t-s)L_k} [\partial_\beta H_k, P_k]\, ds. \tag{41}$$

This formula also holds when paired with $\partial_\alpha H_k$: $\partial_\beta H_k u \in H^1_{\mathrm{per}}$, and the free propagator preserves this form domain.

We now use (38). Since $[\partial_\beta H_k, P_k] = -L_k \partial_\beta P_k$ and $\partial_\beta P_k$ is off-diagonal, integration by parts gives

$$-i \int_0^t s\, e^{-i(t-s)L_k} [\partial_\beta H_k, P_k]\, ds = t \partial_\beta P_k - i(e^{-itL_k} - 1) L_k^+ \partial_\beta P_k.$$

By the Hellmann–Feynman theorem,

$$\begin{aligned} \partial_\alpha \,\mathrm{Tr}(H_k P_k) &= \mathrm{Tr}(\partial_\alpha H_k P_k), \\ \partial_\alpha \partial_\beta \,\mathrm{Tr}(H_k P_k) &= N e_\alpha \cdot e_\beta + \mathrm{Tr}(\partial_\alpha H_k \partial_\beta P_k). \end{aligned}$$

Together with $\partial_\alpha H_{k+\varepsilon t e_\beta} = \partial_\alpha H_k + \varepsilon t e_\alpha \cdot e_\beta$, these identities prove (39). The uniform bounds justify the stated limit interchanges, and Proposition 2.1 removes the equilibrium current. □

5 Proof of the conductivity formula

We prove here Theorem 2.4. We first take the limit $\varepsilon \to 0^+$ in Proposition 4.1, and cancel the term proportional to t before separating the regions near and far from the Dirac points.

For $n \le N < m$, write

$$\begin{aligned} \Delta_{nm,k} &= \lambda_{m,k} - \lambda_{n,k}, \\ z_{nm,k} &= \langle u_{n,k}, \partial_\beta H_k u_{m,k} \rangle \langle u_{m,k}, \partial_\alpha H_k u_{n,k} \rangle. \end{aligned} \tag{42}$$

The finite number of occupied states, the uniform bounds on $\partial_\alpha H_k u_{n,k}$ and $\partial_\beta H_k u_{n,k}$, and Cauchy–Schwarz give

$$\sum_{n\le N<m} |z_{nm,k}| \le C. \tag{43}$$

This controls the infinitely many unoccupied bands. Sums over complete eigenspaces are independent of the choice of basis at internal band crossings.

Let $F(k) = \mathrm{Tr}(H_k P_k)$. The spectral derivative (38) and the eigenbasis representation (36) give

$$\partial_\alpha \partial_\beta F(k) = N e_\alpha \cdot e_\beta - 2 \sum_{n\le N<m} \frac{\mathrm{Re}\, z_{nm,k}}{\Delta_{nm,k}}.$$

The gap is bounded below away from the crossings and is comparable to $|k - k_i|$ near k_i. Thus this derivative is integrable. Applying Stokes formula on the momentum torus with disks of radius r removed gives

$$\int_{\mathcal{B}} \partial_\alpha \partial_\beta F(k)\, dk = 0. \tag{44}$$

Indeed, $\partial_\alpha F(k) = \mathrm{Tr}(\partial_\alpha H_k P_k)$ is bounded, so the extra boundary integrals are $O(r)$ and vanish as $r \to 0$.

For the remaining term of (39), we use

$$\begin{aligned} &\mathrm{Tr}\left(\partial_\alpha H_k (e^{-itL_k} - 1) L_k^+ \partial_\beta P_k\right) \\ &\quad = \sum_{n\le N<m} (e^{it\Delta_{nm,k}} - 1) \frac{z_{nm,k}}{\Delta_{nm,k}^2} - \text{c.c.} \end{aligned}$$

Time reversal leaves the gaps unchanged and conjugates the products $z_{nm,k}$. Their imaginary part therefore integrates to zero. Using (44), we obtain

$$j^{\mathrm{LR}}(t) = \frac{1}{2\pi^2} \int_{\mathcal{B}} \sum_{n\le N<m} \mathrm{Re}\, z_{nm,k} \frac{\sin(t\Delta_{nm,k})}{\Delta_{nm,k}^2}\, dk. \tag{45}$$

The preceding decompositions are legitimate: $|\sin(t\Delta)|/\Delta^2 \le t/\Delta$ and $|1 - \cos(t\Delta)|/\Delta^2 \le t^2/2$, so every term is integrable by (43) and the conical gap bound.

We can now average in time. By Fubini's theorem,

$$\frac{1}{T}\int_0^T j^{\mathrm{LR}}(t)\, dt = \frac{1}{2\pi^2} \int_{\mathcal{B}} \sum_{n\le N<m} \mathrm{Re}\, z_{nm,k} K_T(\Delta_{nm,k})\, dk, \qquad K_T(\Delta) = \frac{1 - \cos(T\Delta)}{T\Delta^3}. \tag{46}$$

This formula makes the large-time behavior explicit: a gap bounded away from zero contributes $O(T^{-1})$, while near a Dirac point the relevant momenta have size T^{-1}.

We introduce a small parameter $\delta > 0$ controlling the size of the neighborhood of the Dirac points, which is independent of T. We decompose $\mathcal{B}$ as the disjoint union

$$\mathcal{B} = \mathcal{B}^\delta_{\mathrm{out}} \cup (\cup_{i=1}^M \mathcal{B}_i^\delta), \qquad \mathcal{B}_i^\delta = B(k_i, \delta),$$

with all sets understood modulo the reciprocal lattice. The radius is chosen small enough that the balls are disjoint and the two crossing bands are separated from the other bands in each ball. We keep δ fixed throughout the proof.

In the following analysis, we first treat the region $\mathcal{B}^\delta_{\mathrm{out}}$, where the gap is bounded below. In the sets $\mathcal{B}_i^\delta$, we study the structure of the Taylor expansion of H_k close to the Dirac points and construct two-band reduced Hamiltonians $H^R_{i,k}$. We reduce successively from H to $H^R_{i,k}$, and finally to the Dirac Hamiltonian

$$H_k^D = v \begin{pmatrix} 0 & k_1 - ik_2 \\ k_1 + ik_2 & 0 \end{pmatrix}, \qquad v > 0,$$

for which we can explicitly compute the limiting current.

5.1 Far from the Dirac points

On $\mathcal{B}^\delta_{\rm out}$, the energy differences $\Delta_{nm,k}$ are bounded away from zero for each fixed δ. Since $|K_T(\Delta)| \leq 2/(T\Delta^3)$, the bound (43) shows that the contribution of this region to (46) is $O_\delta(T^{-1})$. Here O_δ permits a constant depending on the fixed δ.

5.2 Close to Dirac points: reduction to the 2-band case

When $n \neq N$ or $m \neq N+1$, the denominators in (46) are bounded from below in the balls, independently of δ. These transitions also contribute $O(T^{-1})$. Putting the two regions together, we obtain

$$\frac{1}{T}\int_0^T j^{\rm LR}(t)\,dt = \frac{1}{2\pi^2}\sum_{i=1}^M I_i^R(\delta,T) + O_\delta(T^{-1}), \tag{47}$$

where

$$I_i^R(\delta,T) = \int_{\mathcal{B}_i^\delta} \mathrm{Re}\, z_{N,N+1,k}\frac{1-\cos(T\Delta_{N,N+1,k})}{T\Delta^3_{N,N+1,k}}\,dk. \tag{48}$$

At this stage of the proof, only two modes are involved in the formula giving the current, namely the two modes that cross at the Fermi level. Everything happens as for a two-band model that we now study.

5.3 Close to the Dirac points: the local model

We now are interested in the computation of $I_i^R(\delta,T)$. In the following, we drop the index i and assume without loss of generality that $k_i = 0$. We write $v = v_i$.

Hypothesis (26) implies that, for k small enough, the Bloch Hamiltonian H_k has exactly two eigenvalues (counting multiplicities) close to $\mu_{\rm F}$. Consider an arbitrary orthonormal basis (v_0, w_0) of $\mathrm{Ran}(P_{N+1,0} - P_{N-1,0})$. For all k small enough, we can construct an orthonormal basis (v_k, w_k) of $\mathrm{Ran}(P_{N+1,k} - P_{N-1,k})$ by Löwdin orthonormalization of $((P_{N+1,k} - P_{N-1,k})v_0, (P_{N+1,k} - P_{N-1,k})w_0)$, and set

$$H_k^R = [v_k|w_k]^* H_k [v_k|w_k] = \begin{pmatrix} \langle v_k, H_k v_k\rangle & \langle v_k, H_k w_k\rangle \\ \langle w_k, H_k v_k\rangle & \langle w_k, H_k w_k\rangle \end{pmatrix}.$$

It follows that the reduced Hamiltonian

$$H_k\Big|_{\mathrm{Ran}(P_{N+1,k}-P_{N-1,k})}$$

is equivalent through a unitary transform that depends analytically on k to the reduced 2×2 Hamiltonian

$$H_k^R = \sum_{p=0}^3 b^p(k)\sigma_p, \tag{49}$$

where

$$\sigma_0 = \mathrm{Id}_{\mathbb{C}^2},\quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},\quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are the Pauli matrices, and $(b^p)_{p=0,1,2,3}$ are real-valued analytic functions of k in a neighborhood of 0.

The matrix H_k^R has eigenvalues

$$\lambda_\pm(k) = b^0(k) \pm \sqrt{\sum_{p=1}^3 b^p(k)^2}.$$

It follows that

$$b^0(k) = \mu_F + O(|k|^2),\quad b^p(k) = v\langle q^p, k\rangle + O(|k|^2),\quad p = 1,2,3,$$

where the $(q^p)_{p=1,2,3}$ are the rows of a 3×2 matrix Q with orthonormal columns, so that

$$H_k^R = \mu_F + v(Qk) \cdot \sigma + O(|k|^2). \tag{50}$$

Let $R \in \mathrm{SO}(3)$ be a rotation matrix that maps $\mathrm{Ran}(Q)$ to $\mathrm{Span}(e_1^0, e_2^0)$, where (e_1^0, e_2^0, e_3^0) is the canonical basis of $\mathbb{R}^3$. Write R as a rotation through an angle θ about a unit vector n. The unitary matrix

$$U = \cos(\theta/2)\mathrm{Id}_2 - i \sin(\theta/2) n \cdot \sigma$$

satisfies, by the Pauli matrix identities,

$$R_{pq} = \frac{1}{2} \mathrm{Tr}(\sigma_p U \sigma_q U^*).$$

It follows that

$$\begin{aligned} \mathrm{Tr}\big(\sigma_3 U H_k^R U^*\big) &= v \sum_{q=1}^{3} (Qk)_q \, \mathrm{Tr}(\sigma_3 U \sigma_q U^*) + O(|k|^2) \\ &= 2v \langle e_3^0, RQk \rangle + O(|k|^2) \\ &= O(|k|^2). \end{aligned}$$

Up to a unitary transform, we can therefore assume Q to be a 2×2 matrix in (50).

5.4 The two-band case: reduction to the Dirac Hamiltonian

Reduction to H_k^R For $k \neq 0$, let $\lambda_{\pm,k}^R$ be the larger and smaller eigenvalues of H_k^R respectively, and $u_{\pm,k}^R$ associated orthonormal eigenvectors in $\mathbb{C}^2$. We have $\lambda_{-,k}^R = \lambda_{N,k}, \lambda_{+,k}^R = \lambda_{N+1,k}$, and

$$[v_k|w_k] u_{+,k}^R = e^{i\theta_+(k)} u_{N+1,k}, \qquad [v_k|w_k] u_{-,k}^R = e^{i\theta_-(k)} u_{N,k}$$

for some phases $\theta_\pm(k) \in \mathbb{R}$. We have

$$\begin{aligned} \partial_\alpha H_k^R &= [v_k|w_k]^* \partial_\alpha H_k [v_k|w_k] + \partial_\alpha [v_k|w_k]^* [v_k|w_k] H_k^R + H_k^R [v_k|w_k]^* \partial_\alpha [v_k|w_k] \\ &= [v_k|w_k]^* \partial_\alpha H_k [v_k|w_k] + O(|k|) \end{aligned}$$

where we have used for the first line that H_k commutes with $P_{N+1,k} - P_{N-1,k} = [v_k|w_k][v_k|w_k]^*$, and for the second that $H_k^R = \mu_F \mathrm{Id}_2 + O(|k|)$ and $\partial_\alpha \Big([v_k|w_k]^* [v_k|w_k] \Big) = \partial_\alpha \mathrm{Id}_2 = 0$. We therefore obtain

$$\begin{aligned} \langle u_{+,k}^R, \partial_\alpha H_k^R u_{-,k}^R \rangle &= e^{-i(\theta_+(k) - \theta_-(k))} \langle u_{N+1,k}, \partial_\alpha H_k u_{N,k} \rangle + O(|k|) \\ \langle u_{-,k}^R, \partial_\beta H_k^R u_{+,k}^R \rangle &= e^{+i(\theta_+(k) - \theta_-(k))} \langle u_{N,k}, \partial_\beta H_k u_{N+1,k} \rangle + O(|k|). \end{aligned}$$

The phases cancel in $z_{N,N+1,k}$. Replacing the physical velocities by the derivatives of H_k^R therefore changes this product by $O(|k|)$.

Reduction to H_k^D By standard results of perturbation theory [2, Chapter XII] applied to $H_k^R = \mu_F + v(Qk) \cdot \sigma + O(|k|^2)$, with gap greater than a constant multiple of $|k|$,

$$\lambda_{\pm,k}^R = \mu_F + \lambda_{\pm,k}^Q + O(|k|^2), \qquad u_{\pm,k}^R = u_{\pm,k}^Q + O(|k|),$$

where the superscript Q refers to eigenvalues and appropriately chosen orthonormal eigenvectors of the Hamiltonian

$$H_k^Q = v(Qk) \cdot \sigma.$$

Write $\Delta(k) = \Delta_{N,N+1,k}$, $z(k) = z_{N,N+1,k}$, and set

$$z^Q(k) = \langle u_{-,k}^Q, \partial_\beta H_k^Q u_{+,k}^Q \rangle \langle u_{+,k}^Q, \partial_\alpha H_k^Q u_{-,k}^Q \rangle.$$

The preceding estimates give, uniformly in direction,

$$\Delta(k) = 2v|k| + O(|k|^2), \qquad z(k) = z^Q(k) + O(|k|).$$

The function z^Q depends only on the direction of k.

We now make the change of variables $k = q/T$ in (48):

$$I^R(\delta, T) = \int_{\mathbb{R}^2} \mathbb{1}_{\{|q|<T\delta\}} \operatorname{Re} z(q/T) \frac{1-\cos(T\Delta(q/T))}{(T\Delta(q/T))^3}\, dq. \tag{51}$$

For each $q \neq 0$, $T\Delta(q/T) \to 2v|q|$ and $z(q/T) \to z^Q(q)$. Moreover, on the support of the integrand, $c|q| \leq T\Delta(q/T) \leq C|q|$ and $|z(q/T)| \leq C$. Using $1-\cos s \leq \min(s^2/2, 2)$, its absolute value is bounded by

$$C \min\{|q|^{-1}, |q|^{-3}\},$$

which is integrable on $\mathbb{R}^2$. Dominated convergence therefore gives

$$\lim_{T\to\infty} I^R(\delta, T) = I^Q, \qquad I^Q = \int_{\mathbb{R}^2} \operatorname{Re} z^Q(q) \frac{1-\cos(2v|q|)}{(2v|q|)^3}\, dq. \tag{52}$$

This change of scale handles the nonlinear gap inside the oscillatory factor directly.

We perform the change of variable $q' = Qq$ (recall that Q is orthogonal) and obtain

$$I^Q = (Qe_\beta)^T I^D (Qe_\alpha),$$

where the coefficients of the 2×2 matrix I^D are

$$I^D_{ij} = \int_{\mathbb{R}^2} \operatorname{Re}\left(\langle u^D_{-,k}, \partial_{k_i} H^D_k u^D_{+,k}\rangle \langle u^D_{+,k}, \partial_{k_j} H^D_k u^D_{-,k}\rangle\right) \frac{1-\cos(2v|k|)}{(2v|k|)^3}\, dk,$$

and the superscript D refers to the Dirac Hamiltonian

$$H^D_k = vk \cdot \sigma.$$

The Dirac Hamiltonian H^D_k We finish by computing I^D explicitly. Let $k = r(\cos\theta, \sin\theta)$. We have

$$\lambda^D_{\pm,k} = \pm vr, \qquad u^D_{+,k} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ e^{i\theta} \end{pmatrix}, \qquad u^D_{-,k} = \frac{1}{\sqrt{2}} \begin{pmatrix} -e^{-i\theta} \\ 1 \end{pmatrix}.$$

By an explicit calculation, we obtain

$$\int_0^{2\pi} \operatorname{Re}\left(\langle u^D_{-,k}, \partial_{k_i} H^D_k u^D_{+,k}\rangle \langle u^D_{+,k}, \partial_{k_j} H^D_k u^D_{-,k}\rangle\right) d\theta = \pi v^2 \delta_{ij}.$$

It follows that

$$\begin{aligned} I^D_{ij} &= \pi v^2 \delta_{ij} \int_0^\infty \frac{1-\cos(2vr)}{(2vr)^3}\, r\, dr \\ &= \frac{\pi}{4}\delta_{ij} \int_0^\infty \frac{1-\cos s}{s^2}\, ds = \frac{\pi^2}{8}\delta_{ij}. \end{aligned}$$

For the last integral, the identity $s^{-2} = \int_0^\infty a e^{-as}\, da$ and Tonelli's theorem give

$$\int_0^\infty \frac{1-\cos s}{s^2}\, ds = \int_0^\infty a \left(\frac{1}{a} - \frac{a}{a^2+1}\right) da = \frac{\pi}{2}.$$

Since Q is orthogonal, each crossing contributes $\pi^2(e_\alpha \cdot e_\beta)/8$ to the limit of I^R_i. We finally get from (47)

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T j^{\mathrm{LR}}(t)\, dt = \frac{1}{2\pi^2} \frac{M\pi^2}{8}\, e_\alpha \cdot e_\beta = \frac{M}{16}\, e_\alpha \cdot e_\beta,$$

which proves the result.

A Proofs of two technical lemmata

A.1 Proof of Lemma 3.1

Proof. We use the proof of the Faris–Lavine theorem in [1]. It consists in verifying the following two hypotheses of [1, Theorem X.37]. Let $A = -\frac{1}{2}\Delta + W + V$ and $K = A + 2c|x|^2 + b$, where $b \in \mathbb{R}$ will be specified below:

$$\textit{there exists } h, \textit{ such that for any } \phi \in \mathcal{C}, \quad \|A\phi\| \leq h\|K\phi\|; \tag{53}$$

$$\textit{for some } \ell, \textit{ for any } \phi \in \mathcal{C}, \quad |(A\phi, K\phi) - (K\phi, A\phi)| \leq \ell\|K^{\frac{1}{2}}\phi\|^2. \tag{54}$$

By hypothesis *3* in Lemma 3.1 and the conditions on W, it is possible to choose b so that $K \geq 1$. As quadratic forms on $\mathcal{C}$,

$$K^2 = (A+b)^2 + 4c\sum_{j=1}^{2} x_j(A + b + c|x|^2)x_j - 4c.$$

Hypotheses *1* and *3* guarantee that $A + b + c|x|^2$ is bounded below. Hence, increasing the value of b if necessary to make this operator positive, we have

$$\|(A+b)\phi\|_{L^2}^2 \leq \|K\phi\|_{L^2}^2 + 8c\|\phi\|_{L^2}^2,$$

which proves (53).

For (54), we observe that

$$\pm i[A, K] = \pm 2c(x \cdot (-i\nabla) + (-i\nabla) \cdot x) \leq 2c\left((-\Delta) + |x|^2\right) \leq \ell K,$$

where we have used

$$(-\Delta) + |x|^2 \pm (x \cdot (-i\nabla) + (-i\nabla) \cdot x) = (-i\nabla \pm x)^2 \geq 0$$

and

$$K = \left(\frac{a}{2}(-\Delta) + V\right) + (W + c|x|^2) + \frac{1-a}{2}(-\Delta) + c|x|^2 + b \geq e((-\Delta) + |x|^2),$$

where $e = \min(c, \frac{1-a}{2}) > 0$ and where b is chosen so that

$$b - f + \min\,\sigma\left(\frac{a}{2}(-\Delta) + V\right) \geq 0.$$

This proves (54). Hence A is essentially self-adjoint on $\mathcal{C}$. □

A.2 Proof of Lemma 3.2

Proof. By the Kato-Rellich theorem, for any $0 \leq t \leq T$, $H(t)$ is self-adjoint on L^2_{per} with domain H^2_{per}, and bounded below. We will show that there exists $\mu > 0$ so that the graph norm of $(H(t) + \mu)$ for any $0 \leq t \leq T$ is equivalent to the H^2_{per}-norm. Together with the assumed C^1 dependence on t, this verifies the hypotheses of the propagator theorem [1, Theorem X.70], and proves Lemma 3.2.

We have for any $\mu > 0$, $0 \leq t \leq T$ and $\phi \in H^2_{\text{per}}$,

$$\|(H(t) + \mu)\phi\|_{L^2_{\text{per}}} \leq (1+a)\|H_0\phi\|_{L^2_{\text{per}}} + (b+\mu)\|\phi\|_{L^2_{\text{per}}} \leq (1 + a + b + \mu)\|\phi\|_{H^2_{\text{per}}},$$

and so the graph norm is controlled by the H^2_{per}-norm.

For the other inequality, we relate the resolvent of $H(t)$ to that of H_0 by a bounded operator, with bounded inverse. Notice that, for any $\mu > 0$, since H_0 is positive,

$$\forall\, 0 \leq t \leq T, \quad (H(t) + \mu) = (1 + H_1(t)(H_0 + \mu)^{-1})(H_0 + \mu).$$

Furthermore,

$$\forall\, 0 \le t \le T, \quad \|H_1(t)(H_0+\mu)^{-1}\| \le a\|H_0(H_0+\mu)^{-1}\| + b\|(H_0+\mu)^{-1}\| \le a + \frac{b}{\mu}.$$

and so, for $\mu > \frac{b}{1-a}$, the operator $1 + H_1(t)(H_0+\mu)^{-1}$ is bounded and invertible with bounded inverse in L^2_{per}. Therefore $(H(t)+\mu)^{-1}$ is bounded from L^2_{per} to H^2_{per}, which means there exists $C > 0$ such that, for any $\phi \in H^2_{\text{per}}$ and $0 \le t \le T$,

$$\|\phi\|_{H^2_{\text{per}}} = \|(H(t)+\mu)^{-1}(H(t)+\mu)\phi\|_{H^2_{\text{per}}} \le C\|(H(t)+\mu)\phi\|_{L^2_{\text{per}}},$$

which concludes the proof. □

References

[1] Michael C. Reed and Barry Simon. *Methods of Modern Mathematical Physics. II. Fourier analysis, self-adjointness.* Academic Press, New York, 1975.

[2] Michael C. Reed and Barry Simon. *Methods of Modern Mathematical Physics. IV. Analysis of operators.* Academic Press, New York, 1978.